EGR 304 Main Controller Power Budget

Project Name: Main RGB LED Controller
Controller: PIC18F57Q43 Curiosity Nano
Input Supply: 9 VDC, 3 A
Design Safety Margin: 25%

All Major Components

Component Name	Part Number / Design Basis	Supply Voltage Range	Qty.	Current per Unit (mA)	Total Current (mA)	Power (W)	Notes
Red LED Bank Q1	Cree C5SMF-RJE-CU14QBB2	9 V rail	1	83.08	83.08	0.75	4 parallel strings; 3 red LEDs/string; 130 ohm/string
Red LED Bank Q2	Cree C5SMF-RJE-CU14QBB2	9 V rail	1	83.08	83.08	0.75	4 parallel strings; 3 red LEDs/string; 130 ohm/string
Green LED Bank Q3	Cree C5SMF-GJF-CV0Y0791	9 V rail	1	88.00	88.00	0.79	4 parallel strings; 2 green LEDs/string; 100 ohm/string
Green LED Bank Q4	Cree C5SMF-GJF-CV0Y0791	9 V rail	1	88.00	88.00	0.79	4 parallel strings; 2 green LEDs/string; 100 ohm/string
Blue LED Bank Q5	Cree C5SMF-BJE-CR0U0451	9 V rail	1	88.00	88.00	0.79	4 parallel strings; 2 blue LEDs/string; 100 ohm/string
PIC18F57Q43 Curiosity Nano	Microchip DM164150	5 V rail	1	100.00	100.00	0.50	Conservative design allowance; not a measured board current
Indicator LEDs	Design allowance	5 V rail	3	20.00	60.00	0.30	Temp LED 1, Temp LED 2, Trouble LED; conservative 20 mA each
L7805 quiescent current	ST L7805CV	9 V input	1	6.00	6.00	0.05	Conservative regulator operating-current allowance

+9 V Power Rail

Component Name	Part Number / Design Basis	Supply Voltage Range	Qty.	Current per Unit (mA)	Total Current (mA)	Power (W)	Notes
RGB LED Array	5 MOSFET-controlled banks	9 V	1	430.16	430.16	3.87	Maximum nominal array current at 100% PWM
5 V Regulator Branch	L7805 input current allowance	9 V	1	166.00	166.00	1.49	160 mA 5 V load + ~6 mA regulator allowance
				Subtotal	**596.16**		
				Safety Margin	**0.25**		
				Total Current Required on +9 V Rail	**745.20**	**6.71**	

+5 V Regulated Rail

Component Name	Part Number / Design Basis	Supply Voltage Range	Qty.	Current per Unit (mA)	Total Current (mA)	Power (W)	Notes
PIC18F57Q43 Curiosity Nano	Microchip DM164150	5 V	1	100.00	100.00	0.50	Conservative design allowance
Temperature LED 1	Design allowance	5 V	1	20.00	20.00	0.10	Worst-case indicator allowance
Temperature LED 2	Design allowance	5 V	1	20.00	20.00	0.10	Worst-case indicator allowance
Trouble LED	Design allowance	5 V	1	20.00	20.00	0.10	Worst-case indicator allowance
				Subtotal	**160.00**		
				Safety Margin	**0.25**		
				Total Current Required on +5 V Rail	**200.00**	**1.00**	

Regulator / Supply Capacity

Component Name	Part Number / Design Basis	Supply Voltage Range	Qty.	Current per Unit (mA)	Total Current (mA)	Power (W)	Notes
5 V Regulator	ST L7805CV	7-35 V input / 5 V output	1	1500.00	1500.00	0.64	Regulator current rating; thermal dissipation must also be respected
9 V DC Supply	External adapter	9 VDC	1	3000.00	3000.00	27.00	Available source current
F1 Mini Blade Fuse	Littelfuse 0297002.WXNV	32 VDC max	1	2000.00	2000.00		Final selected fuse rating
				Remaining 9 V Supply Current (with 25% margin)	**2254.80**		
				L7805 Estimated Dissipation at 160 mA load		**0.64**	
				Recommended Main Power Trace	**>= 1.02 mm (40 mil), 2 oz Cu**		
				Preferred Jack-to-Fuse Feed if space permits	**1.5-2.0 mm**		

Design Notes / Sources

Cree LED datasheet	/mnt/data/HB-C5SMF-RJF-RJE-GJF-GJE-BJF-BJE(1).pdf
PIC Nano documentation	/mnt/data/PIC18F57Q43-Curiosity-Nano-HW-UserGuide-DS40002186B.pdf
Peralta PCB mill specifications	/mnt/data/Printed Circuit Boards Mill Specs _ Peralta Engineering Studio.pdf
Power-budget basis	RGB 430.16 mA + regulator branch 166 mA = 596.16 mA; 25% margin = 745.20 mA